

Admin

HW 2 9/25 11:59AM

- OH room change (Univ 320) ✓
- HW1 grades returned ✓
- Practice midterm ✓

Last time

- Introduced multivariate r.v.s (discrete and continuous)
- Functions of continuous bivariate r.v.s

Independent r.v.s

Recall: $A, B \subset S$ are independent if $P(AB) = P(A)P(B)$ (Independence of events)

DEF: Two random variables X and Y are independent if for any two sets

$C \subseteq \mathbb{R}, D \subseteq \mathbb{R}$, we have that

$$P(X \in C, Y \in D) = P(X \in C)P(Y \in D) \quad (\text{Independence of r.v.s})$$

Independence of discrete r.v.s: If X and Y are discrete, then they are independent if and only if $p_{XY}(x, y) = p_X(x)p_Y(y)$ for all $x, y \in \mathbb{R}$

means X, Y independent $\Rightarrow p_{XY}(x, y) = p_X(x)p_Y(y)$

AND $p_{XY}(x, y) = p_X(x)p_Y(y) \Rightarrow X, Y$ independent

Independence of continuous r.v.s: If X, Y are continuous, then they are independent if and only if $f_{XY}(x, y) = g(x)h(y) \quad \forall x, y \in \mathbb{R}$
for some functions g, h $\Rightarrow$ "for all"

$\Rightarrow$ Example: $f(x, y) = \begin{cases} 6e^{-2x}e^{-3y} & \text{for } 0 < x < \infty, 0 < y < \infty \\ 0 & \text{otherwise} \end{cases}$

Are X, Y independent? $f(x, y) = 6e^{-2x}e^{-3y} \mathbb{1}_{\{0 < x < \infty, 0 < y < \infty\}}$
 $\text{Yes } \checkmark$
 $= 6e^{-2x}e^{-3y} \mathbb{1}_{\{0 < x < \infty\}} \mathbb{1}_{\{0 < y < \infty\}}$
 $= \underbrace{6e^{-2x} \mathbb{1}_{\{0 < x < \infty\}}}_{g(x)} \times \underbrace{e^{-3y} \mathbb{1}_{\{0 < y < \infty\}}}_{h(y)}$

Example: $f(x, y) = \begin{cases} 24xy & \text{for } 0 < x < 1, 0 < y < 1, 0 < x+y < 1 \\ 0 & \text{otherwise} \end{cases}$

Are X, Y independent? **No** $f(x, y) = 24xy \mathbb{1}_{\{0 < x < 1, 0 < y < 1, 0 < x+y < 1\}}$
 $= 24xy \mathbb{1}_{\{0 < x < 1\}} \mathbb{1}_{\{0 < y < 1\}} \mathbb{1}_{\{0 < x+y < 1\}}$

cannot be factored
into separate functions
of x and y
 $\Rightarrow$ not independent

Example: A man and a woman agree to meet. Each person independently arrives at some time uniformly distributed between 2PM and 3PM. What is the probability that the man arrives at least 10 minutes before the woman?

Hint: Let X = # of minutes past 2PM when the man arrives $X \sim \text{Unif}(0, 60)$
 (Step 0) Y = # of minutes past 2PM when the woman arrives $Y \sim \text{Unif}(0, 60)$

Step 1: Write down the prob. you need to calculate

$$P(X < Y - 10) = P(X + 10 < Y)$$

Step 2: Integrate $f_{X,Y}(x, y) = f_X(x)f_Y(y) = \left(\frac{1}{60}\right)^2 \mathbb{1}_{\{0 < x < 60, 0 < y < 60\}}$

$$\int \int_{x < y-10} \left(\frac{1}{60}\right)^2 dx dy = \int_{10}^{60} \int_0^{y-10} \left(\frac{1}{60}\right)^2 dx dy \dots$$

$$\int \int_{x < y-10} \left(\frac{1}{60}\right)^2 dy dx = \int \int_{x+10 < y} \left(\frac{1}{60}\right)^2 dy dx = \int_0^{50} \int_{10+x}^{60} \left(\frac{1}{60}\right)^2 dy dx \dots$$

Convolution: What is the distribution of the sum of independent r.v.s?

THM: If X and Y are independent, continuous r.v.s with pdfs f_X and f_Y respectively, then the pdf of $Z = X + Y$ is

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(z-y)f_Y(y) dy$$

We say f_Z is the convolution of f_X and f_Y , represented as

$$f_Z = f_X \star f_Y$$

↘ Convolution

Proof: Step 1: Compute the cdf of Z

$$F_Z(z) = P(Z \leq z) = P(X+Y \leq z)$$

$$= \int \int_{x+y \leq z} f_{X,Y}(x, y) dx dy$$

$$= \int \int_{x+y \leq z} f_X(x)f_Y(y) dx dy \quad (\text{Independence})$$

$$= \int \int_{x \leq z-y} f_X(x)f_Y(y) dx dy$$

$$= \int \int_{-\infty}^{z-y} f_X(x)f_Y(y) dx dy$$

$$= \int_{-\infty}^{\infty} f_Y(y) \left(\int_{-\infty}^{z-y} f_X(x) dx \right) dy$$

$$= \int_{-\infty}^{\infty} f_Y(y) F_X(z-y) dy$$

$$\begin{aligned} F_X(z-y) &= P(X \leq z-y) \\ &= \int_{-\infty}^{z-y} f_X(x) dx \end{aligned}$$

Step 2: Compute $f_z(z)$

$$f_z(z) = \frac{d}{dz} F_z(z) \stackrel{\text{Step 1}}{=} \frac{d}{dz} \int_{-\infty}^{\infty} f_Y(y) F_X(z-y) dy$$

$$= \int_{-\infty}^{\infty} f_Y(y) \frac{d}{dz} F_X(z-y) dy \quad f_X(z-y) \times \frac{d}{dz} z$$

$$= \int_{-\infty}^{\infty} f_Y(y) f_X(z-y) dy \quad \text{Done!}$$

Example: Suppose X, Y are independent r.v.s and both are $\text{Unif}(0,1)$
What is the pdf of $Z = X + Y$?

$$f_z(z) = \int_{-\infty}^{\infty} f_Y(y) f_X(z-y) dy$$

$$= \int_{-\infty}^{\infty} \mathbb{1}_{\{0 < y < 1\}} \mathbb{1}_{\{0 < z-y < 1\}} dy$$

$$= \int_0^1 \mathbb{1}_{\{0 < z-y < 1\}} dy$$

$$\vdots$$

$$= (1 - |z-1|) \quad (\text{ex. ...})$$

$$X \sim \text{Unif}(0,1)$$

$$f_X(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ 1 & \text{if } 0 < x < 1 \\ 0 & \text{if } x \geq 1 \end{cases}$$

$$= \mathbb{1}_{\{0 < x < 1\}}$$

Conditional distributions

e.g., (X, Y) are (height, weight)

Discrete r.v.s

DEF: Let X and Y be discrete r.v.s with joint pmf $p_{XY}(x, y)$ and marginal pmfs $p_X(x)$ and $p_Y(y)$. For any y where $p_Y(y) > 0$, the conditional pmf of X given $Y = y$ is denoted by $p_{X|Y}(x|y)$ is defined as

$$p_{X|Y}(x|y) = P(X = x | Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)} = \frac{p_{XY}(x, y)}{p_Y(y)}$$

Similarly, if $p_X(x) > 0$, then

$$p_{Y|X}(y|x) = P(Y = y | X = x) = \frac{p_{XY}(x, y)}{p_X(x)}$$

Note: If $A, B \subset S$ then $P(B|A)$ is still a "valid" probability in the sense that it satisfies the 3 axioms

Similarly, the conditional pmf is a "valid" pmf

$$(1) p_{X|Y}(x|y) \geq 0 \quad \forall x$$

$$(2) \sum_x p_{X|Y}(x|y) = \sum_x \frac{p_{XY}(x, y)}{p_Y(y)} = \frac{1}{p_Y(y)} \sum_x p_{XY}(x, y) = \frac{p_Y(y)}{p_Y(y)} = 1 \quad \checkmark$$

Independence: If X and Y are independent then

$$P_{X|Y}(x|y) = \frac{P_{XY}(x,y)}{P_Y(y)} \downarrow = \frac{P_X(x)P_Y(y)}{P_Y(y)} = P_X(x)$$

$$\equiv P(X=x|Y=y) = P(X=x)$$

DEF: The conditional distribution function of X given $Y=y$ is denoted by $F_{X|Y}(x|y)$ and is given by

$$F_{X|Y}(x|y) = P(X \leq x | Y=y) \text{ for } y \text{ s.t. } P(Y=y) > 0$$

Example:

	$X=0$	$X=1$
$Y=0$	0.4	0.1
$Y=1$	0.2	0.3

(a) Find $P_X(x)$

$$\text{Compute } P_X(0) = P(X=0) = P(X=0, Y=0) + P(X=0, Y=1) = 0.4 + 0.2 = 0.6$$

$$P_X(1) = P(X=1) = P(X=1, Y=0) + P(X=1, Y=1) = 0.1 + 0.3 = 0.4$$

(b) Find $P_{X|Y}(x|1) = P(X=x|Y=1)$

$$P_{X|Y}(0|1) = P(X=0|Y=1) = \frac{P(X=0, Y=1)}{P(Y=1)} = \frac{P(X=0, Y=1)}{P(X=0, Y=1) + P(X=1, Y=1)}$$
$$= \frac{0.2}{0.2 + 0.3}$$

$$P_{X|Y}(1|1) = \frac{0.3}{0.5}$$

Continuous r.v.s

DEF: Let X and Y be continuous r.v.s with joint density $f_{XY}(x,y)$ and marginal densities $f_X(x)$ and $f_Y(y)$, respectively. For any y such that $f_Y(y) > 0$, the conditional pdf of $X|Y=y$, written $f_{X|Y}(x|y)$ is defined by

$$f_{X|Y}(x|y) = \frac{f_{XY}(x,y)}{f_Y(y)} \quad \left\{ \begin{array}{l} f_X(x) > 0 \\ f_{Y|X}(y|x) = \frac{f_{XY}(x,y)}{f_X(x)} \end{array} \right.$$

Exercise: verify that $f_{X|Y}(x|y)$ is a valid density

Independence: If X and Y are independent

$$f_{X|Y}(x|y) = \frac{f_{XY}(x,y)}{f_Y(y)} = \frac{f_X(x)f_Y(y)}{f_Y(y)} = f_X(x)$$

Example: Suppose (X, Y) has joint density

$$f(x, y) = \begin{cases} \frac{21}{4} x^2 y & (0 \leq) x^2 \leq y < 1 \\ 0 & \text{otherwise} \end{cases} \quad \text{implicit}$$

(a) Compute $f_{X|Y}(x|y) = \frac{f_{XY}(x, y)}{f_Y(y)}$

$$\begin{aligned} f_Y(y) &= \int_{-\sqrt{y}}^{\sqrt{y}} \frac{21}{4} x^2 y \, dx = \frac{21}{4} y \int_{-\sqrt{y}}^{\sqrt{y}} x^2 \, dx = \frac{21}{4} y \times \left[\frac{1}{3} x^3 \right]_{-\sqrt{y}}^{\sqrt{y}} \\ &= \frac{21}{4} y \left(\frac{1}{3} y^{3/2} - \frac{1}{3} (-y)^{3/2} \right) \\ &= \frac{7}{4} y (2 y^{3/2}) = \frac{7}{2} y^{5/2} \quad \text{if } 0 < y < 1 \end{aligned}$$

$$f_{X|Y}(x|y) = \frac{\frac{21}{4} x^2 y}{\frac{7}{2} y^{5/2}} = \frac{3}{2} x^2 y^{-3/2} \quad \text{if } x^2 \leq y \quad (\text{or } -\sqrt{y} \leq x \leq \sqrt{y})$$

(b) Compute $P(X \geq Y | Y = y) = \int_{x \geq y} \frac{3}{2} x^2 y^{-3/2} \, dx$

$$= \int_y^1 \frac{3}{2} x^2 y^{-3/2} \, dx = \dots$$

$$f_{Y|X=t}(y|t)$$